Type MS · Rates of change with one chain-rule link

Answer 1 of 18 · Question 2 part whole · 6 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

2	$\frac{dA}{dt} = 8$	B1
	$\frac{dA}{dr} = 2\pi r$	M1
	$A = 50 \quad r = \sqrt{\frac{50}{\pi}}$ (3.989...)	M1
	$\frac{dr}{dt} = \frac{dr}{dA} \times \frac{dA}{dt}, = \frac{1}{2\pi\sqrt{\frac{50}{\pi}}} \times 8, = 0.319$ (cm/s)	M1,A1ft,A1 [6]
NB	For either method, accept A or S for area, r for radius. Any other letters used for area and/or radius must be defined.	
B1	$\frac{dA}{dt} = 8$ seen explicitly or used	
M1	Attempt to differentiate πr^2 to obtain $\frac{dA}{dr}$ Power of r must decrease	
M1	Attempt to obtain r when $A = 50$ cm^2 (ie solve $50 = \pi r^2$)	
M1	For a correct, useful, chain rule. Derivatives can appear in any order	
A1ft	Substitute their known quantities and rearrange to $\frac{dr}{dt} = ...$ if not in this form already. **All 3 M marks needed**	
A1	Correct answer, **must be 3 sf**	
ALT	$\frac{dA}{dt} = 8$	B1
	$r = \sqrt{\frac{a}{\pi}}$ oe	M1
	$\frac{dr}{dA} = \frac{1}{2\sqrt{\pi}} A^{-\frac{1}{2}}$	M1
	$\frac{dA}{dt} \times \frac{dr}{dA} = \frac{dr}{dt}$	M1
	$= 8 \times \frac{1}{2\sqrt{\pi}} A^{-\frac{1}{2}} = 8 \times \frac{1}{2\sqrt{\pi}} \times \frac{1}{\sqrt{50}}$	A1ft
	$= 0.3191... = 0.319$ (cm/s)	A1
B1	$\frac{dA}{dt} = 8$ seen explicitly or used	
M1	Attempt to find r in terms of A	
M1	Attempt to differentiate their expression for r to obtain $\frac{dr}{dA}$ power of A must decrease	
M1	For a correct, useful, chain rule. Derivatives can appear in any order	
A1ft	Substitute their known quantities and rearrange to $\frac{dr}{dt} = ...$ if not in this form already. All 3 M marks needed	
A1	Correct answer, **must be 3 sf**	

Type MS · Rates of change with one chain-rule link

Answer 1 of 18 · Question 2 part whole · 6 marks

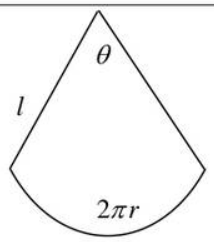
2019 JUN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{dy}{}$	

Type MS · Rates of change with one chain-rule link

Answer 2 of 18 · Question 10 part b · 3 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

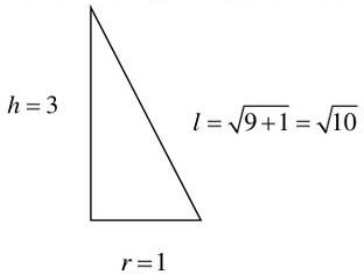
Number		
10(a)	Circumference of base = $2\pi r$ 	B1
	$l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	B1
	$A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l} = \pi rl$	M1A1 (4)
(b)	$A = \pi rl$	
	$l = r\sqrt{10} \Rightarrow A = \pi r^2\sqrt{10}$	B1
	$\dfrac{dA}{dr} = 2\pi r\sqrt{10} \Rightarrow k = 2\sqrt{10}$	M1A1 (3)
(c)	$\dfrac{dV}{dt} = 1.5\ \left(\text{cm}^3/\text{s}\right)$	B1
	$\dfrac{dA}{dt} = \dfrac{dA}{dr} \times \dfrac{dr}{dV} \times \dfrac{dV}{dt}$	M1
	$V = \dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi r^2 \times 3r = \pi r^3$	B1
	$\dfrac{dV}{dr} = 3\pi r^2$	B1ft
	$\therefore \dfrac{dA}{dt} = 2\pi \times 8\sqrt{10} \times 1.5 \times \dfrac{1}{3\times 64\pi} = 0.3952... = 0.395\ \text{cm}^2/\text{s}$	A1 (5)
		[12]

Part	Mark	Notes	
(a)	B1	For the circ of the base $L = 2\pi r$	
	B1	$R = l$ and $l = r\theta$ Therefore $l\theta = 2\pi r \Rightarrow \theta = \dfrac{2\pi r}{l}$	$R = l$
	M1	$A = \dfrac{1}{2}l^2\theta$ and substituting their expression for θ to give $A = \dfrac{1}{2}l^2\theta = \dfrac{1}{2}l^2\dfrac{2\pi r}{l}$	Uses the formula $A = \dfrac{1}{2}RL$ $A = \dfrac{1}{2}\times l \times 2\pi r \Rightarrow \left(A = \pi rl\right)$
	A1	For the required expression for A, $A = \pi rl$ with no errors.	

Type MS · Rates of change with one chain-rule link

Answer 2 of 18 · Question 10 part b · 3 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

(b)	B1	For finding that the slant height is $\sqrt{10}$ times the radius of the cone  So $l=r\sqrt{10}$
	M1	Substitutes $l=r\sqrt{10}$ into the given expression $A=\pi rl$ and differentiates their resulting expression to find $\frac{dA}{dr}$ $A=\pi r^2\sqrt{10}$ so $\frac{dA}{dr}=2\pi r\sqrt{10}$
	A1	Therefore $k=2\sqrt{10}$
(c)	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dr}\times\frac{dr}{dV}\times\frac{dV}{dt}$
	B1	For finding the volume of a cone in terms of r only $V=\frac{1}{3}\pi r^2h=\frac{1}{3}\pi r^2\times 3r=\pi r^3$
	B1ft	Differentiates their expression for the volume of a cone provided it is in terms of V and r only. $\frac{dV}{dr}=3\pi r^2$
	A1	For combining all required terms into their chain rule and evaluating to 3 significant figures, $\frac{dA}{dt}=2\pi\times 8\sqrt{10}\times 1.5\times\frac{1}{3\times 64\pi}=0.3952...=0.395\ \text{cm}^2/\text{s}$
	ALT – in terms of h	
	B1	States $\frac{dV}{dt}=1.5\ \left(\text{cm}^3/\text{s}\right)$ Award if it seen explicitly in (b) **and used** in (c)
	M1	States (or uses) a correct chain rule $\frac{dA}{dt}=\frac{dA}{dh}\times\frac{dh}{dt}$ and $\frac{dV}{dt}=\frac{dV}{dh}\times\frac{dh}{dt}$
	B1	For finding the area and volume of a cone in terms of h $r=\frac{h}{3}$, $A=\pi r^2\sqrt{10}\Rightarrow A=\frac{\sqrt{10}}{9}\pi h^2$ and $V=\frac{1}{3}\pi r^2h\Rightarrow V=\frac{1}{27}\pi h^3$
	B1ft	Differentiates their expressions for the area and volume of a cone provided they are both in terms of h only. $\frac{dA}{dh}=\frac{2\sqrt{10}}{9}\pi h$ and $\frac{dV}{dh}=\frac{3}{27}\pi h^2$
	A1	Combines the required terms into their chain rules and evaluating to 3 significant figures $\frac{dA}{dt}=0.395\ 102$

Type MS · Rates of change with one chain-rule link

Answer 2 of 18 · Question 10 part b · 3 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	1	

Type MS · Rates of change with one chain-rule link

Answer 3 of 18 · Question 6 part whole · 7 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

Question	Scheme	Marks
6	$\frac{dV}{dt}=12$	B1
	$\frac{dV}{dh}=9h^2$	B1
	$1536=3h^3 \Rightarrow h=\sqrt[3]{\frac{1536}{3}}=8$	M1A1
	$\frac{dh}{dt}=\frac{dV}{dt}\times\frac{1}{\frac{dV}{dh}}=12\times\frac{1}{9h^2}=12\times\frac{1}{9\times8^2}=\frac{1}{48}$ (cm/s) oe	M1dM1A1
		[7]
		Total 7 marks

Mark	Notes
B1	For stating $\frac{dV}{dt}=12$ This must be clearly labelled $\frac{dV}{dt}$
B1	For differentiating the **given** expression for the volume. This must be clearly labelled $\frac{dV}{dh}$ **It must be correct for this mark.**
M1	For using the given formula, rearranged correctly to find the height of oil when the volume = 1536 That is, unless you see $h = 8$, you must see this expression. $h=\sqrt[3]{\frac{1536}{3}}$
A1	For the correct value of $h = 8$ Sight of $h = 8$ without working is M1A1
M1	For sight of a **correct** chain rule involving $\frac{dh}{dt}, \frac{dV}{dh}$ and $\frac{dV}{dt}$ only. Accept in any order. For example, accept $\frac{dh}{dt}\times\frac{dV}{dh}=\frac{dV}{dt}$ that is, $\frac{dh}{dt}$ does not need to be the subject. This mark can be implied by a correct next step.
dM1	For substituting in their values/expressions with their value of h into a correct chain rule. Note, this mark is dependent on the previous M mark.
A1	For the correct rate of increase. This question asks for an exact value. Do not accept a decimal estimate or accept $0.0208\dot{3}$ unless you see a recurring sign.

Type MS · Rates of change with one chain-rule link

Answer 3 of 18 · Question 6 part whole · 7 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

	Accept any fraction that simplifies to $\frac{1}{48}$. You will see $\frac{12}{576}$ which is completely acceptable.

Question	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 4 of 18 · Question 7 part whole · 6 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

Number		
7	$\frac{dV}{dt} = -k$	B1
	$\left\{\frac{dV}{dh}\right\} = 40\pi h - \pi h^2$	M1
	$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = -\frac{k}{40\pi h - \pi h^2}$	M1A1
	$-\frac{1}{60} = -\frac{k}{40\times\pi\times12 - \pi\times12^2}\left\{= -\frac{k}{336\pi}\right\} \Rightarrow k = \frac{28}{5}\pi\left(\text{cm}^3/\text{s}\right)$	dM1A1 [6]
		Total 6 marks

Part	Mark	Additional Guidance
	B1	For identifying that $\frac{dV}{dt} = -k$ seen/embedded in a correct chain rule, accept $\pm k$
	M1	Attempt to find $\frac{dV}{dh}$, condone if $\frac{dV}{dh}$ is not present or incorrect If attempted by expanding the bracket first, they need to achieve $Ph \pm Qh^2$ If attempted by product rule, score for $Ah(60-h) \pm Bh^2$
	M1	A correct chain rule formed by using $\frac{dV}{dh}, \frac{dh}{dt}$ and $\frac{dV}{dt}$ (in any order) stated or implied
	A1	Obtains the correct chain rule having at least two terms with values/expressions substituted in, no errors seen. e.g. $-k = \left(40\pi h - \pi h^2\right) \times \frac{dh}{dt}$ or $\frac{dV}{dt} = \left(40\pi h - \pi h^2\right) \times -\frac{1}{60}$
	dM1	Substitutes $h = 12, \frac{dV}{dt} = \pm k, \frac{dh}{dt} = \pm\frac{1}{60}$ and their $\frac{dV}{dh} = '40\pi h - \pi h^2'$ to a correct chain rule to find $k = ...$ Dependent on the 2nd M mark only
	A1	Correct answer only for using correct signs throughout $\frac{28}{5}\pi$ or 5.6π, ignore unit

Type MS · Rates of change with one chain-rule link

Answer 4 of 18 · Question 7 part whole · 6 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 5 of 18 · Question 3 part whole · 6 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

3	Let l be the length. $\frac{dl}{dt} = 0.005$	
	$l = 10x \Rightarrow \frac{dl}{dx} = 10$ or $\frac{dl}{dt} = 10\frac{dx}{dt} = 0.005 \Rightarrow \frac{dx}{dt} = 0.0005$	B1
	$r = \frac{x}{2}, \quad V = \pi\left(\frac{x}{2}\right)^2 \times 10x \Rightarrow V = \frac{5\pi x^3}{2}$	B1
	$\frac{dV}{dx} = \frac{15\pi x^2}{2}$	M1
	$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dl} \times \frac{dl}{dt}$ or $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt}$	M1
	$\frac{dV}{dt} = \frac{15\pi \times 3^2}{2} \times \frac{1}{10} \times 0.005$ or $\frac{dV}{dt} = \frac{15\pi \times 3^2}{2} \times 0.0005$ $= 0.106028...$	dM1
	$\frac{dV}{dt} \approx 0.11 \text{ (cm}^3\text{/s)}$	A1 [6]
Alternative		
	$x = \frac{l}{10}$ or $r = \frac{l}{20}$ stated or used	B1
	$V = \pi\left(\frac{l}{20}\right)^2 \times l = \frac{\pi l^3}{400}$	B1
	$\frac{dV}{dl} = \frac{3\pi l^2}{400}$	M1
	$\frac{dV}{dt} = \frac{dV}{dl} \times \frac{dl}{dt}$	M1
	$\frac{dV}{dt} = \frac{3\pi \times 30^2}{400} \times 0.005 = 0.106028...$	dM1
	$\frac{dV}{dt} \approx 0.11 \text{ (cm}^3\text{/s)}$	A1 [6]
		Total 6 marks

Type MS · Rates of change with one chain-rule link

Answer 5 of 18 · Question 3 part whole · 6 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Apart from x, which is defined in the question, different variables may be used providing that they are consistent throughout the working. Score the best mark using either alternative but not a mixture of both.		
	B1	$\frac{dl}{dx}=10$ or $\frac{dx}{dt}=0.0005$
	B1	Correct expression for the volume of the cylinder. Simplification not needed. Brackets for $\left(\frac{x}{2}\right)^2$ are required unless implied by subsequent working.
	M1	Attempt to differentiate to find $\frac{dV}{dx}$. You need to see a multiple of x^2 from differentiating x^3. ft their V if it is a multiple of x^3. Mark the differentiation and ignore any error in the label, $\frac{dV}{dx}$, for this mark only.
	M1	State, or use correctly, the chain rule $\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dl}\times\frac{dl}{dt}$ or $\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dt}$
	dM1	Substitute to find a numerical expression for $\frac{dV}{dt}$ dep on both M marks. ft their $\frac{dV}{dx}, \frac{dl}{dx}, \frac{dx}{dt}$ if clearly stated.
	A1	$\frac{dV}{dt}\approx 0.11$ **given to 2SF**

Alternative		
	B1	$x=\frac{l}{10}$ or $r=\frac{l}{20}$ stated or used
	B1	Correct expression for the volume of the cylinder. Simplification not needed. Brackets for $\left(\frac{l}{20}\right)^2$ are required unless implied by subsequent working.
	M1	Attempt to differentiate to find $\frac{dV}{dl}$. You need to see a multiple of l^2 from differentiating l^3. ft their V if it is a multiple of l^3. Mark the differentiation and ignore any error in the label, $\frac{dV}{dl}$, for this mark only.
	M1	State, or use correctly, the chain rule $\frac{dV}{dt}=\frac{dV}{dl}\times\frac{dl}{dt}$
	dM1	Substitute to find a numerical expression for $\frac{dV}{dt}$ dep on both M marks. ft their $\frac{dV}{dl}$ if clearly stated.
	A1	$\frac{dV}{dt}\approx 0.11$ **given to 2SF**

Type MS · Rates of change with one chain-rule link

Answer 5 of 18 · Question 3 part whole · 6 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 6 of 18 · Question 8 part a · 3 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number		
8 (a)	$4\times1.5=6$	B1
	$\pi r^2=6$	M1
	$r=\sqrt{\frac{6}{\pi}}$	A1
		(3)
(b)	$\frac{dA}{dr}=2\pi r$	M1
	$\frac{dr}{dt}=\frac{dr}{dA}\times\frac{dA}{dt}\left[=\frac{1}{2\pi r}\times1.5\right]$	M1
	$=\frac{1}{2\pi\sqrt{\frac{6}{\pi}}}\times1.5$	M1
	$=0.173$	A1
		(4)
		Total 7 marks

Type MS · Rates of change with one chain-rule link

Answer 6 of 18 · Question 8 part a · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 7 of 18 · Question 8 part b · 4 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
8 (a)	$4\times1.5=6$	B1
	$\pi r^2=6$	M1
	$r=\sqrt{\frac{6}{\pi}}$	A1
		(3)
(b)	$\frac{dA}{dr}=2\pi r$	M1
	$\frac{dr}{dt}=\frac{dr}{dA}\times\frac{dA}{dt}\left[=\frac{1}{2\pi r}\times1.5\right]$	M1
	$=\frac{1}{2\pi\sqrt{\frac{6}{\pi}}}\times1.5$	M1
	$=0.173$	A1
		(4)
		Total 7 marks

Type MS · Rates of change with one chain-rule link

Answer 7 of 18 · Question 8 part b · 4 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 8 of 18 · Question 6 part whole · 6 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

6	$A = \frac{r^2}{2} \times \frac{\pi}{6} = \left(\frac{\pi r^2}{12}\right)$	B1
	$\frac{dA}{dr} = \frac{2 \times \pi r}{12} = \left(\frac{\pi r}{6}\right)$	M1
	$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$	M1
	Length of arc $AB = \frac{5\pi}{2} = r\frac{\pi}{6} \Rightarrow r = 15$	M1
	$\frac{dA}{dt} = \frac{\pi \times 15}{6} \times 0.2 = \frac{\pi}{2}\ \left(\text{cm}^2/\text{s}\right)$	M1A1 [6]
		Total 6 marks

Question	Notes	Marks
6	For using the correct formula for the area of the sector $A = \frac{r^2}{2} \times \frac{\pi}{6} = \left(\frac{\pi r^2}{12}\right)$	B1
	For an attempt to differentiate their expression for the area provided it is in the form $A = kr^2$ where $k \neq 1$ (see general guidance) $\frac{dA}{dr} = \frac{2 \times \pi r}{12} = \left(\frac{\pi r}{6}\right)$	M1
	For a correct statement of chain rule to achieve $\frac{dA}{dt}$ $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$ may be seen stated or implied by substitution.	M1
	For using the length of arc when the length of arc is $\frac{5\pi}{2}$ $\frac{5\pi}{2} = r\frac{\pi}{6} \Rightarrow (r = 15)$	M1
	For attempting to find the rate of change of area using a correct chain rule, their $\frac{dA}{dr}$, the given $\frac{dr}{dt}$ and their value for r $\frac{dA}{dt} = \frac{'\pi'}{6} \times '15' \times 0.2 = \ldots$	M1
	For the correct value of $\frac{dA}{dt} = \frac{\pi}{2}\ \left(\text{cm}^2/\text{s}\right)$	A1 [6]
		Total 6 marks

Type MS · Rates of change with one chain-rule link

Answer 8 of 18 · Question 6 part whole · 6 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 9 of 18 · Question 4 part whole · 8 marks

2024 JUN · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

Question	Scheme	Marks
4	$\frac{dA}{dt}=50\pi \quad \frac{dr}{dt}=\frac{5}{12}$	B1,B1
	$\left(A=4\pi r^2 \Rightarrow \frac{dA}{dr}=\right)8\pi r$	B1
	$\frac{dr}{dt}=\frac{1}{\frac{dA}{dr}}\times\frac{dA}{dt}$ **oe** $\Rightarrow \frac{5}{12}=\frac{1}{8\pi r}\times 50\pi \Rightarrow r=15$	M1dM1A1
	$V=\frac{4}{3}\times\pi\times 15^3=4500\pi\left[\text{cm}^3\right]$	M1A1
		[8]
		Total 8 marks

Mark	Notes
B1	For either the correct $\frac{dA}{dt}=50\pi$ or $\frac{dr}{dt}=\frac{5}{12}$ May be seen explicitly or used implicitly in the candidate's work.
B1	For both correct $\frac{dA}{dt}=50\pi$ and $\frac{dr}{dt}=\frac{5}{12}$ May be seen explicitly or implicitly in the candidate's working.
B1	For $8\pi r$ May be seen explicitly or implicitly in the candidate's working.
M1	For a correct chain rule, relevant to the question. Look for equivalences. This may be explicitly stated or can be awarded for use of the appropriate values or expressions implicitly.
dM1	For correctly substituting their values and rearranging their equation to find a value for r. Allow errors in rearrangement. Dependent on the previous method mark, though M1 dM1 can be awarded if the implicit use of their values in a correct chain rule is correct, without the chain rule being stated.
A1	For r = 15 (cm)
M1	For using the formula for the volume of a sphere with their r **Note – although this has not been made a dependent mark, this mark can only be awarded if 'their r' has come from some attempt at calculus.**
A1	For the correct volume of a sphere given exactly.

Type MS · Rates of change with one chain-rule link

Answer 9 of 18 · Question 4 part whole · 8 marks

2024 JUN · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 10 of 18 · Question 5 part whole · 6 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

5	$\frac{dr}{dt} = 0.7$	B1
	$\{\frac{dF}{dr}\} = -2 \times \frac{3}{20} r^{-3}$ oe	M1A1
	$\{\frac{dF}{dt} = \frac{dF}{dr} \times \frac{dr}{dt} =\} -\frac{0.3}{2.8^3} \times 0.7 = -0.0095663 \ldots$ (newtons/s) Accept **awrt** -0.00957 or -9.57×10^{-3}	M1dM1A1 [6]
		Total 6 marks

Mark	Notes
B1	For writing $\frac{dr}{dt} = 0.7$ can be implied or embedded in later work. Condone using letter ***d*** or ***D*** or ***R*** only instead of ***r.***
M1	For an attempt to differentiate the formula for F, for $\ldots r^{-2} \rightarrow \ldots r^{-3}$ oe Attempts quotient rule, look for $\frac{-Pr}{Qr^4}$ $(P > 0, Q > 0)$ oe, if the quotient rule is quoted, it must be correct, if clearly they use incorrect quotient rule M0
A1	For a correct differentiated expression simplified or unsimplified
M1	For a **correct** chain rule, accept in any order. (correct chain rule stated or implied by their working) Condone using letter d instead of r. For example, accept $\frac{dF}{dr} = \frac{dF}{dt} \times \frac{dt}{dr}$ *or* $\frac{dF}{dr} = \frac{dF}{dt} \div \frac{dr}{dt}$ $\left(\frac{dF}{dt}\right.$ does not need to be the subject$\left.\right)$
dM1	For substituting the value of $r=2.8$ into their dF/dr in a **correct** chain rule, leading to $\frac{dF}{dt} = \ldots$.(a value, not a numerical expression) , the value can be a fraction or decimal. If their final answer is not correct, explicit substitution must be seen. **Depends on the previous method mark only.**
A1	For the correct value of $\frac{dF}{dt}$ (Condone lack of units or incorrect units) isw

Type MS · Rates of change with one chain-rule link

Answer 10 of 18 · Question 5 part whole · 6 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 11 of 18 · Question 9 part a · 5 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

9 (a)	$\frac{dA}{dt} = 0.03$	B1
	$A = \frac{1}{2}x^2 \sin 60^\circ = \frac{\sqrt{3}}{4}x^2$	M1
	$\frac{dA}{dx} = \frac{\sqrt{3}}{2}x$	A1
	When $x = 2$ $\frac{dx}{dt} = \frac{1}{\sqrt{3}} \times 0.03 = 0.0173$ cm/s	M1 A1 (5)
(b)	$V = \sqrt{3}x^3$ $\frac{dV}{dx} = 3\sqrt{3}x^2$	M1
	When $x = 2$ $\frac{dV}{dt} = 12\sqrt{3} \times 0.0173 = 0.36$	M1 A1 (3)
		Total 8 marks

Type MS · Rates of change with one chain-rule link

Answer 11 of 18 · Question 9 part a · 5 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Guidance
(a)	B1	For stating or using correctly in their Chain Rule $\frac{dA}{dt}=0.03$
	M1	For using the **correct** formula $\left(\frac{1}{2}ab\sin C\right)$ with the correct lengths and angle of 60° or $\frac{\pi}{3}$, for the cross-sectional area of the prism to obtain $A=\frac{1}{2}x^2\sin 60^\circ=\left(\frac{\sqrt{3}}{4}x^2\right)$ and differentiating their expression which must be as a minimum $A=px^2$ to obtain $\frac{dA}{dx}=qx$ [where p and q are constants]. [The height of the triangle is $\frac{\sqrt{3}}{2}x$ if they use $\frac{1}{2}\times\text{base}\times\text{height}$]
	A1	For the correct $\frac{dA}{dx}=\frac{\sqrt{3}}{2}x$
	M1	For applying a **correct** Chain rule using their $\frac{dA}{dx}$ and $x=2$ to obtain $\frac{dx}{dt}=\left(\frac{1}{\frac{dA}{dx}}\times\frac{dA}{dt}\right)=\frac{dx}{dA}\times\frac{dA}{dt}=\left(\frac{2}{\sqrt{3}}\right)\times\frac{1}{2}\times 0.03$
	A1	$\frac{dx}{dt}=0.0173$
(b)	M1	For a correct expression for the volume using their A from part (a) to obtain $V=\frac{\sqrt{3}}{4}x^2\times 4x=\left(\sqrt{3}x^3\right)$ and differentiating their expression which must be as a minimum $V=mx^3$ to obtain as a minimum $\frac{dV}{dx}=nx^2$ [where m and n are constants] $\left(\frac{dV}{dx}=3\sqrt{3}x^2\right)$
	M1	For applying a correct Chain rule using their $\frac{dV}{dx}$ and $x=2$ to obtain $\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dt}=12\sqrt{3}\times 0.0173=\lceil 0.359...\rceil$ $\left(\text{ft their }\frac{dx}{dt}\right)$ **Note:** $\frac{dx}{dt}=0.0173$ or $\frac{\sqrt{3}}{100}$ $\left(\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dt}=12\sqrt{3}\times\frac{\sqrt{3}}{100}=\frac{9}{25}=0.36\right)$
	A1	For awrt 0.36

Type MS · Rates of change with one chain-rule link

Answer 11 of 18 · Question 9 part a · 5 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 12 of 18 · Question 6 part b · 4 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

6 (a)	$\left[V = \frac{1}{3}\pi r^2 h\right]$ $\tan 30^\circ = \frac{r}{h} \Rightarrow r = \frac{h}{\sqrt{3}}$ $V = \frac{1}{3}\pi\left(\frac{h}{\sqrt{3}}\right)^2 h \Rightarrow V = \frac{1}{9}\pi h^3$ *	M1 A1 cso [2]
(b)	$\frac{dV}{dh} = \frac{\pi h^2}{3}$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} \Rightarrow \frac{dh}{dt} = \frac{3}{\pi h^2} \times -0.9 = \frac{3}{\pi \times 1.2^2} \times -0.9 = -0.59683...$ ≈ -0.597 cm/s [Accept an answer of ± 0.597 cm/s]	M1 M1M1 A1 [4]
		Total 6 marks

Type MS · Rates of change with one chain-rule link

Answer 12 of 18 · Question 6 part b · 4 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 13 of 18 · Question 6 part whole · 6 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

	mark scored.
A1	$x = 6$ Must reject $x = -4$ if this solution is also included.

Type MS · Rates of change with one chain-rule link

Answer 13 of 18 · Question 6 part whole · 6 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks
6	$\left[\frac{dV}{dt} = 3\ (\text{cm}^3/\text{s})\right]$ $V = \frac{4}{3}\pi r^3 \quad \frac{dV}{dr} = 4\pi r^2, \quad A = 4\pi r^2 \quad \frac{dA}{dr} = 8\pi r$ $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dV} \times \frac{dV}{dt}$ oe $\frac{dA}{dt} = [8\pi \times 10] \times \left[\frac{1}{4\pi \times 10^2}\right] \times 3 = 0.6\ (\text{cm}^2/s)$	M1A1,A1 (M1 for any one) M1 dM1A1 [6]
		Total 6 marks

Type MS · Rates of change with one chain-rule link

Answer 14 of 18 · Question 2 part whole · 6 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

2	$(V=)3x^3$	B1
	$\frac{dV}{dx}=9x^2$	M1
	$\left(\frac{dx}{dt}=\right)\frac{dV}{dt}\times\frac{dx}{dV}$ or $\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dt}$ oe	M1
	$\left(\frac{dx}{dt}=\right)\frac{8}{9x^2}$ oe	A1
	$\left(\frac{dx}{dt}=\right)\frac{8}{9\times2^2}$ oe	dM1
	$\frac{2}{9}$ oe	A1 [6]
		Total 6 marks

Type MS · Rates of change with one chain-rule link

Answer 14 of 18 · Question 2 part whole · 6 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 15 of 18 · Question 6 part b · 5 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks
6 (a)	$\left(V=\right)5\times\frac{1}{2}\times r^2\times\sin\left(\frac{\pi}{3}\right)=\frac{5\sqrt{3}}{4}r^2$ *	B1*cso [1]
(b)	$\left(\frac{\mathrm{d}V}{\mathrm{d}r}=\right)\frac{5\sqrt{3}}{2}r$ oe	M1
	(When the area of $BCDF = 60\ \text{cm}^2$, $BC = DF = r$ =)12 cm	B1
	$\frac{\mathrm{d}V}{\mathrm{d}t}=\frac{\mathrm{d}V}{\mathrm{d}r}\times\frac{\mathrm{d}r}{\mathrm{d}t}$ oe	M1
	$\left(\frac{\mathrm{d}V}{\mathrm{d}t}=\right)\frac{5\sqrt{3}}{2}\times12\times0.2=6\sqrt{3}$ oe $\left(\text{cm}^3/\text{s}\right)$	dM1A1 [5]
		Total 6 marks

Type MS · Rates of change with one chain-rule link

Answer 15 of 18 · Question 6 part b · 5 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 16 of 18 · Question 5 part whole · 5 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

5	$\left\{\frac{dV}{dh}\right\} 18h^2$	M1
	$\frac{dV}{dt} = (\pm)36$	B1
	$384 = 6h^3 \Rightarrow h^3 = 64$ or $h = 4$	B1
	$\left\{\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}\right\}$	
	$\left\{\frac{dh}{dt}\right\} = \frac{1}{18 \times 4^2} \times (\pm)36$ oe	M1
	$= -\frac{1}{8}$ oe	A1
		Total 5 marks

Type MS · Rates of change with one chain-rule link

Answer 16 of 18 · Question 5 part whole · 5 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 17 of 18 · Question 7 part a · 4 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Number		
7 (a)	$(V=)x^3 \quad (\frac{dV}{dx}=)3x^2 \quad (\text{at } x=2 \quad \frac{dV}{dx}=12)$	M1
	$\frac{dx}{dt}=0.1$	B1 (A1 on ePen)
	$\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dt}="12"\times 0.1$ oe	M1
	1.2 m^3/s cao oe	A1cao (4)
(b)	(Surface Area =) $6x^2 \quad (\frac{dA}{dx})=12x \quad$ at $x=6$	M1
	$\frac{dA}{dx}=72$	A1
	$\frac{dV}{dx}=108 \quad \frac{dA}{dt}=0.05$	B1
	$\frac{dV}{dt}=\frac{dV}{dx}\times\frac{dx}{dA}\times\frac{dA}{dt}="108"\times"\frac{1}{72}"\times 0.05$	M1
	0.075 m^3/s	A1 (5)
	ALT	
	A = $6x^2$ leading to an expression in A for $V \quad V=\left(\frac{A}{6}\right)^{\frac{3}{2}}$	M1
	$\frac{dV}{dA}=\frac{1}{4}\left(\frac{A}{6}\right)^{\frac{1}{2}}$ oe	A1
	$\frac{dV}{dA}=\frac{3}{2}$ and $\frac{dA}{dt}=0.05$ oe	B1
	$\frac{dV}{dt}=\frac{dV}{dA}\times\frac{dA}{dt}="\frac{3}{2}"\times 0.05$	M1
	1.2 m^3/s cao oe	A1cao [5] **[9]**

Type MS · Rates of change with one chain-rule link

Answer 17 of 18 · Question 7 part a · 4 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Correct expression for Volume, attempt at differentiation to ax^2, a is an integer, $a > 1$.
	B1	$\frac{dx}{dt} = 0.1$, can be explicit or implicitly used in a chain rule.
	M1	For **any correct chain rule**, that would lead to a value for $\frac{dV}{dt}$ and substitution of 0.1 and their value for $\frac{dV}{dx}$. They must show an attempt to find $\frac{dV}{dx}$, need not be a correct attempt, this isn't a dependent mark.
	A1	cao oe
(b)	M1	Correct expression for Surface Area, attempt at differentiation to bx, b is an integer, $b > 1$.
	A1	$\frac{dA}{dx} = 72$
	B1	$\frac{dV}{dx} = 108$ & $\frac{dA}{dt} = 0.05$ clearly stated, implicitly in chain rule or explicitly
	M1	For **any correct chain rule**, that would lead to a value for $\frac{dV}{dt}$ and substitution of 0.05 and their values for $\frac{dA}{dx}$ and $\frac{dV}{dx}$ They must show an attempt to find $\frac{dV}{dx}$, need not be a correct attempt, this isn't a dependent mark.
	A1	cao oe
(b) ALT	M1	Correct expression for area, attempt to rearrange, expression for V in terms of A
	A1	oe
	B1	Both derivatives clearly stated, implicitly in a chain rule or explicitly
	M1	For **any correct chain rule**, that would lead to a value for $\frac{dV}{dt}$ and substitution of 0.05 and their value for $\frac{dV}{dA}$. $\frac{dV}{dA}$ doesn't need to come from correct working, but there must have been some attempt to find an expression for V in terms of A and $\frac{dV}{dA}$ presented somewhere in the working.
	A1	cao oe

Type MS · Rates of change with one chain-rule link

Answer 17 of 18 · Question 7 part a · 4 marks

2020 NOV · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Rates of change with one chain-rule link

Answer 18 of 18 · Question 11 part whole · 8 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

11	$\frac{3}{r}=\frac{4}{h}$ oe $\rightarrow$ $r=\frac{3h}{4}$ oe	M1 A1
	$(V=)\frac{1}{3}\pi\left(\frac{3h}{4}\right)^2 h$ $\left(V=\frac{3\pi h^3}{16}\right)$ oe	M1
	$\left(\frac{dV}{dh}=\right)\frac{9\pi h^2}{16}$ oe	M1 A1
	$\frac{dh}{dt}=\frac{dV}{dt}.\frac{dh}{dV}$ oe	M1
	$\left(\frac{dh}{dt}=\right)9\pi\times\frac{16}{9\pi(2)^2}$ oe	dddM1
	$\left(\frac{dh}{dt}=\right)4$	A1 [8]
	OR	
	$\frac{3}{r}=\frac{4}{h}$ oe $\Rightarrow$ $h=\frac{4r}{3}$ oe	M1A1
	$(V=)\frac{1}{3}\pi r^2 h$ $\left(V=\frac{4\pi r^3}{9}\right)$ oe	M1
	$\left(\frac{dV}{dr}=\right)\frac{4\pi r^2}{3}$ AND $\frac{dh}{dr}=\frac{4}{3}$	M1A1
	$\frac{dh}{dt}=\frac{dV}{dt}.\frac{dr}{dV}.\frac{dh}{dr}$ oe	M1
	$\left(\frac{dh}{dt}=\right)9\pi\times\frac{1}{\frac{4\pi\left(\frac{3}{2}\right)^2}{3}}\times\frac{4}{3}$ oe $\left[h=2,\quad r=\frac{3}{2}\right]$	M1
	$\left(\frac{dh}{dt}=\right)4$	A1 [8]
		Total 8 marks

Type MS · Rates of change with one chain-rule link

Answer 18 of 18 · Question 11 part whole · 8 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Additional Guidance
M1	For any correct equation using similar triangles to link together r and h
A1	For $r=\frac{3h}{4}$. This may be implied in working.
M1	For using their expression for r in terms of h to get an expression for V of the form $a\pi h^3$
M1	For a minimally acceptable attempt to differentiate their expression for V to get an equation of the form $b\pi h^2$
A1	For $\left(\frac{dV}{dh}=\right)\frac{9\pi h^2}{16}$ oe
M1	For a correct, relevant chain rule. This may be implied in working.
dddM1	Correct substitution into chain rule, dependent on the previous 3 method marks.
A1	Correct value.
	ALT
M1	For any correct equation using similar triangles to link together r and h
A1	For $h=\frac{4r}{3}$
M1	For using their expression for r in terms of h to get an expression for V of the form $a\pi r^3$
M1	For a minimally acceptable attempt to differentiate their expression for V to get an equation of the form $b\pi r^2$
A1	For $\left(\frac{dV}{dr}=\right)\frac{4\pi r^2}{3}$ AND $\frac{dh}{dr}=\frac{4}{3}$
M1	For a correct, relevant chain rule. This may be implied in working
dddM1	Correct substitution into chain rule, dep on previous three marks
A1	Correct value

Type MS · Rates of change with one chain-rule link

Answer 18 of 18 · Question 11 part whole · 8 marks

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